

Executive Summary

The **Digital Library of Indic Intellectual History (DLIIH)** will create a world-class open-access platform for the earliest layers of Indian thought, adopting a strict top-down chronological approach. The first phase (Year 1–2) will establish a deep vertical slice of the Rigveda Samhita, prioritizing the technical resolution of complex philological challenges including flawless Unicode encoding of Vedic accents (svaras) and the alignment of Samhitapatha with Padapatha. Leveraging modern NLP/AI for automated sandhi splitting and morphological tagging, DLIIH will provide a richly annotated, interoperable corpus. Subsequent phases will extend this foundational architecture to the broader Vedic, Sramana, and classical darshana corpora. Key outputs include a curated Vedic dataset, cross-canonical analysis tools, and an AI-driven translation workflow. Funding will combine grants with a scalable B2B revenue stream—licensing highly structured Vedic data to AI companies for specialized LLM training—to ensure long-term sustainability.

| **Project Name:** Digital Library of Indic Intellectual History (DLIIH)
| **Focus:** Early Sanskrit corpus (Vedas, Brahmanas, Aranyakas, Upanishads)
| **Duration:** 5 years (2026–2031)
| **Budget (est.):** ₹150–350 Lakh total (3 scenarios)
| **Lead:** [Institution/Consortium], India (Project Manager: *TBD*)
| **Partners:** Muktabodha IRI, BORI Pune, IF Pondicherry, GRETIL, SanskritLibrary, IGNCA, IIT Kanpur, key universities, temples/manuscript libraries (see Partners table)
| **Deliverables:** 20+ digitized texts (Veda books & appendices), TEI XML edition & translations, search/index API, AI research assistant, scholarly annotations, educational courses.
| **Tech:** UTF-8 Unicode (incl. Vedic accents)【4†L43-L47】, TEI XML schema, OCR + manual proofreading, Elasticsearch, LLM/RAG for semantic search.
| **Access Model:** Free core content; advanced tools (AI queries, analytics) via subscriptions/grants; donations/crowdfunding.

Background & Need

Vedic literature (ṛg-, yajur-, sāma-, atharva-veda plus Brahmanas, Aranyakas, Upaniṣads) forms the bedrock of ancient Indian thought【8†L88-L97】. However, these texts remain scattered and under-digitized. Existing initiatives (GRETIL, TITUS, Digital Corpus of Sanskrit, SARIT, Muktabodha, SANSKNET, Ambuda, etc.) have made inroads but have gaps: most Vedic content is available only as unmarked plain text or isolated PDFs【10†L179-L187】【10†L199-L205】. For example, GRETIL offers extensive Sanskrit texts but with only basic markup【10†L179-L187】; the Muktabodha Digital Library (an important Sanskrit archive) has ~3,000 manuscripts and 500+ searchable e-texts (mostly Shaiva/Tantric) and only recently began including Vedic material【46†L55-L59】【46†L77-L83】. Government portals (e.g. India’s Vedic Heritage Portal) compile resources and link to digitized volumes【29†L369-L373】，but do not offer unified

search or academic tools. Volunteer projects like Ambuda are rapidly publishing Sanskrit texts and emphasizing open licensing【53†L103-L111】【53†L179-L184】, yet they are general Sanskrit and not specialized for Vedic studies.

This fragmentation limits scholarship. Scholars lack a single platform with standardized encoding (e.g. TEI) and semantic tools (concept tagging, graph links) for comparative Vedic study. Digitization is also incomplete: e.g. the BORI/Pune Rigveda collection (UNESCO Memory of World) was slated for digitization by India's Manuscripts Mission【41†L251-L257】, but many manuscripts remain only in images. Moreover, advanced search across Sanskrit and translations is still nascent. The **DLIIH** addresses these gaps by **integrating and enhancing** existing resources under a common data schema, adding missing texts, and providing modern digital humanities infrastructure (TEI/XML editions, morphological indexing, parallel texts, cross-database links). This will transform access: researchers can, for example, query all passages on *ātman* across Vedic layers and see Sanskrit, translations, and commentary in one view.

Key evidence/gaps: Recent reviews note that digital Vedic projects often lack rich semantic annotation and rely on plain text repositories【10†L179-L187】【10†L190-L194】. There is a need for standardized encoding (the NSF-funded Sanskrit Library project pioneered Unicode/Vedic encoding standards【4†L11-L19】【4†L43-L47】), and for a collaborative platform (e.g. the USA Digital Sanskrit Library prototype at Brown) to build on. DLIIH will leverage these standards (Unicode with Vedic extensions, TEI guidelines) and adopt best practices from digital philology (OCR → transcription → markup). It will be community-driven (open contributions, peer review) and align with efforts like Muktabodha's and Ambuda's by **free licensing all outputs**, ensuring maximum reuse.

Objectives and Scope

Primary Goal: Build a sustainable, open-access digital library that underpins later Indic intellectual traditions. The project adopts a strict chronological priority, focusing first on the Rigveda Samhita as a deep vertical slice to establish a robust technical foundation. The initial scope (Year 1–2) covers:

- **Rigveda Samhita (Deep Vertical Slice):** Implementation of a technical foundation prioritizing philological accuracy, specifically flawless Unicode encoding of Vedic accents (svaras), structural alignment of Samhitapatha with Padapatha, and the encoding of early commentaries such as Sayana-bhasya.
- **Expansion to Vedic Corpus:** Following the Rigveda pilot, the architecture will scale to include the remaining Samhitas (Yajur, Sama, Atharva), core Brahmanas (Aitareya, Satapatha), and Aranyakas.
- **Principal Upanisads:** Integration of early Upanisadic texts to trace the transition from ritual to philosophical inquiry within the same interoperable TEI/XML framework.
- **Principal Upaniṣads:** Early Upaniṣads (e.g. Bṛhadāraṇyaka, Chāndogya, Īśā, Kaṭha, Kena, Kṛṣṇa-Yajur). These form the transition to Jñāna-kāṇḍa.

Secondary scope (Year 2–5) will sequentially add:

- **Early Śramaṇa Texts:** Selected early Buddhist suttas (e.g. Sutta Nipāta, Dhammapada, early Sutta Piṭaka) and Jain āgamas (e.g. Ācārāṅga, Uttarādhyayana), to parallel Vedic metaphysics and ethics.
- **Classical Darśanas:** Major Sūtra/Śāstra works of Nyāya, Vaiśeṣika, Sāṃkhya-Yoga, Mīmāṃsā, Vedānta; and key Buddhist/Jain philosophical texts.
- **Epics & Purāṇas:** Sanskrit epics (Mahābhārata, Rāmāyaṇa) and principal Purāṇas, focusing on their Vedic foundations.
- **Tantric & Bhakti Texts:** A later expansion might include mainstream Tantric scriptures and medieval Bhakti/Sant works to trace intellectual continuities.

However, **Phase I is strictly Vedic** to establish robust encoding standards and workflows. All content (Sanskrit + translations + metadata) will be freely licensed (e.g. CC BY/CC BY-SA or public domain) to encourage use and remix. We will develop the platform architecture from the start to accommodate adding new traditions in later phases, with the *user-facing narrative* organized chronologically (Vedic → Classical → etc) rather than sectarian divisions. This appeals to broad audiences (academics, practitioners) and aligns with Government/academic missions of cultural preservation.

Implementation Plan

A multi-year phased plan will guide development. Each phase has clear deliverables, milestones, and success metrics. A summary timeline and deliverables is below; **Table 1** provides details.

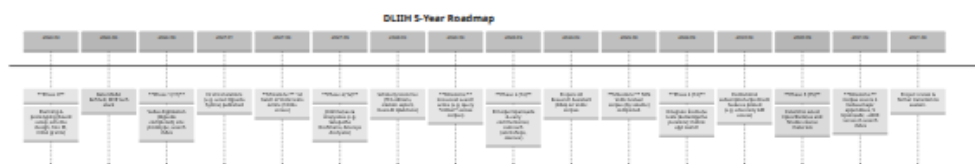


Table 1. Phases, Deliverables & Metrics. (see next page)

Phase	Timeline	Key Deliverables	Milestones / Metrics
Phase 0: Planning & Infrastructure	Q1–Q2 2026	Governance setup; data model & TEI schema design; base infrastructure	Schema published; Beta site live; grant applications submitted

Phase	Timeline	Key Deliverables (cloud, Git, DB); hiring core team	Milestones / Metrics
Phase 1: Vedic Samhitas	Q3 2026–Q2 2027	Complete encoding (OCR+proofread+markup) of R̥gveda, Śukla & Kṛṣṇa Yajurveda, Sāmaveda, Atharvaveda; transliteration; basic English commentary notes; search index	100% texts scanned & OCR-corrected; Searchable; 1st translations (rigveda) online; Usage by pilot scholars
Phase 2: Brahmanas & Āraṇyakas	Q3 2027–Q4 2027	Encode Śatapatha Brāhmaṇa, Āitareya Brāhmaṇa; Aitareya, Taittirīya, Śāṅkhyāyana, Talavakāra Āraṇyakas; links to Saṃhitās; ontology tagging (ritual terms)	5 Brāhmaṇa/Āraṇyaka texts online; citation tools; External academic references to site in progress
Phase 3: Upaniṣads & Early Jñānavarga	Q1–Q4 2028	Encode Brhadāraṇyaka, Chāndogya, Īśa, Kaṭha, Kena, etc. Upaniṣads; introductory commentaries; cross-links (e.g. Āraṇyaka-Upaniṣad); advanced search (semantic tag search); launch	10+ Upaniṣads available; active user base grows (target: 5000 users/month); first courses/conference sessions held

Phase	Timeline	Key Deliverables	Milestones / Metrics
		of educational courses/workshops on the portal.	
Phase 4: Śramaṇa texts	Q1 2029–Q4 2029	Add key Buddhist sūtras (early Pāli and Āgama parallels) and Jain āgamas; align common concepts (e.g. <i>anātmā</i> , <i>karma</i>); implement AI-powered study assistant (RAG system for query-answering).	100 Buddhist/Jain discourses in corpus; AI assistant beta with QA accuracy >80%; Institutional pilot usage (e.g. select universities)
Phase 5: Classical & Outreach	Q1 2030–Q2 2031	Expand to Classical Sanskrit (selected Darśana texts, Epics, Purāṇas); finalize multi-language UI (Hindi, Sanskrit); partner integration (printing/POD books); launch membership courses; refine sustainability (donor program, subscriptions).	Corpus spans Vedic to Classical (~50,000 verses); 10,000+ monthly active users; 200 active volunteers/contributors; Break-even or reserve fund achieved.

Each phase will have internal milestones (e.g. percent of texts encoded, system demos) and success metrics (text volume, search queries, academic citations, user feedback). Regular stakeholder reviews will ensure alignment with goals. Phase transitions depend on funding and volunteer availability, so risks (see below) are managed by overlapping activities.

Technical Architecture & Standards

The platform utilizes a modular stack where AI/NLP serves as the core editorial engine. This automation is critical for handling the unique complexity of archaic Sanskrit grammar at scale.

- **Encoding Standards:** All texts encoded in Unicode UTF-8 using Devanāgarī with Vedic extensions (per Unicode 5.2+【4†L43-L47】) and standardized transliteration (IAST) for cross-lingual search. We will adopt **TEI/XML** guidelines for marking text structure (verses, paragraph breaks, editorial notes) and metadata (scripture identifiers, tradition, manuscripts). This follows digital philology best practices【4†L11-L19】【6†L55-L63】.
- **Data Schema:** A canonical text schema will underpin the corpus (see ER diagram below). Key entities: *Text* (a named scripture/work), *Manuscript/Edition* (provenance info), *Verse/Segment*, *Translation*, *Person* (authors, translators, commentators), *Concept/Tag*, *Commentary*, and *Institution*. Every verse/segment has a stable ID (e.g. Rigveda 1.1.1) and attributes (language, meter, ritual category). The schema supports relationships (e.g. commentary-of, equivalence across traditions).
- **Storage & Search:** Texts and metadata stored in PostgreSQL (JSONB fields for TEI metadata) and indexed in Elasticsearch (for full-text and semantic queries). Semantic search will use multilingual embeddings (HuggingFace or custom models) to retrieve conceptually similar passages (e.g. *dharma*, *ritual*). An API will expose search, text, and metadata endpoints for integration and external use.
- **AI & RAG:** We will develop a Retrieval-Augmented Generation assistant trained on the corpus. Researchers can ask natural-language questions (e.g. “Where is *brahman* discussed as world-soul?”) and the system will return synthesised answers citing verses. This builds on recent Sanskrit RAG projects【37†L0-L4】. A smaller dedicated LLM (fine-tuned on Sanskrit-English parallel data) will ensure accuracy.
- **Tools:** We will integrate existing Sanskrit tools (Sandhi splitters, morphological analyzers like the Digital Corpus of Sanskrit) into workflows. OCR will use state-of-the-art Indic OCR engines (e.g. Google/OCR-D) with manual correction. Version control (Git/GitLab) will track text editions, translations, and site code.
- **AI Editorial Engine:** Machine Learning models will be utilized to automate high-complexity editorial tasks: sandhi splitting to resolve euphonic combinations, morphological tagging of archaic Vedic grammar, and the structural alignment of base verses with classical commentaries and modern translations.

Interface & Design

The platform moves beyond the limitations of "static digital books" by implementing a modern Human-Computer Interaction (HCI) model designed for deep philological work. The interface features a layered, deep-dive architecture with dynamic, split-screen views. This allows scholars to toggle seamlessly between metrical analysis (Chandas), word-by-word grammatical breakdowns, and multiple parallel translations in a single responsive environment.

Below is a proposed mermaid **Entity-Relationship Diagram** for the data model, illustrating major tables and relations:

```
erDiagram
    TEXT {
        string text_id PK
        string title
        string tradition
        string language
        int year (approx)
    }
    MANUSCRIPT {
        int ms_id PK
        string source
        string script
        int century
    }
    EDITION {
        int ed_id PK
        string editor
        date pub_date
        string license
    }
    SEGMENT {
        int seg_id PK
        string text_id FK
        int number
        text content_original
        text content_translit
    }
    TRANSLATION {
        int trans_id PK
        int seg_id FK
        string translator
        text content_en
    }
    COMMENTARY {
        int comm_id PK
        int seg_id FK
        string author
    }
```

```

    text content_note
}
PERSON {
    int  pid PK
    string name
    string role (e.g. author, editor)
}
TEXT ||--o{ EDITION : has
EDITION ||--o{ SEGMENT : contains
SEGMENT ||--o{ TRANSLATION : has
SEGMENT ||--o{ COMMENTARY : has
MANUSCRIPT ||--|| EDITION : source_of
PERSON ||--o{ EDITION : "created by"
PERSON ||--o{ TRANSLATION : "translated by"

```

In practice, **Every verse** in a *TEXT* (e.g. “Rigveda 1.1.1”) is a record in *SEGMENT*, linked to its parent Text and Edition. Translations and commentaries are attached to segments. Manuscripts are linked as sources for editions (metadata only).

Translation & Editorial Workflow

We will establish a rigorous pipeline for each new text: **Scan → OCR → Proofread → Encoding → Translation → Peer Review → Publish.**

1. **Scanning/OCR:** Collaborate with manuscript libraries to scan documents. Use advanced OCR (trained on Devanāgarī and Grantha/Vatteluttu for Malayalam rituals) for initial text.
2. **Proofreading:** Volunteers and paid staff will correct OCR. We will use a web interface (like [SanskritDocs proofreading](#) or customized tools) to crowdsource verification. Mandatory double-key entry or peer review ensures >99% accuracy.
3. **TEI Encoding:** Texts are encoded in XML/TEI: markup for verses, prose, sandhi splits, diacritics (vedic accent, ṁ). Custom tags handle Vedic specifics (e.g. meter, chandas) and rituals (agraha). This preserves both content and scholarly apparatus. We will provide style guides and templates.
4. **Translation:** Each text gets at least a preliminary English translation. We will engage academic collaborators and qualified volunteers to draft translations. A wiki-style platform (or Git workflows) will allow iterative improvement. Glossaries of key terms will be shared.
5. **Peer Review:** Subject experts (Indologists, Vedic scholars) will review translations and encoding. A versioning system with review statuses (Draft → Reviewed → Published) will maintain quality. A clear erratum mechanism will allow corrections post-publication.
6. **Publication:** Once approved, texts and translations are published on the site. Each verse segment gets a permalink and citation (with DOI). Versions are archived, with changelogs. Data will be licensed (likely CC BY-SA) and machine-readable for reuse.

Quality Assurance: The editorial workflow emphasizes accuracy (dual checks), consistency (style guides, controlled vocabularies for concepts), and transparency (version history, open reviews). A QA team will spot-check random samples; annual audits will ensure compliance with TEI and license policies.

Staffing & Governance

DLIIH will operate under a non-profit (or academic consortium) with a lean core team and volunteer network. **Org Structure:**

- **Executive Director/PI** (1 FTE): Oversees strategy, partnerships, fundraising, administrative governance.
- **Technical Lead/CTO** (1 FTE): Architecture design, infrastructure, manages developers and data engineers.
- **Developers/Engineers** (2–3 FTE): Full-stack developers for web, backend, AI/RAG integration, database management.
- **Textual Scholars** (2 FTE): Sanskritists/Linguists to oversee encoding, annotation, coordinate volunteers.
- **Translation Editors** (2 FTE): Manage translation teams, ensure consistency and review.
- **Project Manager / Coordinator** (1 FTE): Handles timelines, communications, volunteer coordination, documentation.
- **Outreach/Education Coordinator** (0.5–1 FTE): Manages courses, partnerships, donor relations, community events.
- **Admin & Finance** (0.5 FTE): Accounting, HR, compliance (could be shared or part-time).

Additionally, we will form an **Advisory Board** of eminent scholars (Indology, Digital Humanities, religious studies) and a **Technical Advisory Committee** (IT librarians, digital preservation experts) to guide policies and mitigate risks.

Crucially, DLIIH will encourage **academic and volunteer collaboration:**

- **Scholarly Network:** We will partner with Sanskrit depts (e.g. JNU, SOAS, U. of Washington), Indology institutes (IFP Pondicherry, CNRS in Paris), and Societies (AVISTA) to involve PhD students and faculty in transcription and translation projects.
- **Volunteer Community:** Similar to Ambuda, we will open contributions via a vetting platform. Verified volunteers (students, retired teachers, practitioners) can proofread text, contribute translations, or annotate with citations. Badges/acknowledgments will incentivize.
- **Institutional Collaborations:** We will formalize agreements with manuscript libraries (BORI, IF Pondicherry, etc.) for data sharing; with publishers/universities for co-hosted courses; and with tech partners (e.g. IIT Kanpur for morphological tools) to strengthen capacity.

Governance: The non-profit entity will adhere to transparent policies (open source code, open meetings for major decisions). Legal counsel will ensure compliance in licensing and privacy. Financial oversight (audits) will maintain donor trust.

Budget Estimate

A comprehensive budget covers personnel, technical infrastructure, digitization, outreach, and contingency. Below is a **line-item estimate** in INR for three scenarios: *Low, Medium, High* funding. These are 5-year cumulative totals.

- **Personnel:** Major cost. Scenarios assume 6–9 full-time staff plus part-time/admin (mid: ~₹120 L/year total; low: ₹80 L; high: ₹200 L).
- **Infrastructure:** Cloud servers, storage, bandwidth (search indexing, AI compute). Annual (\$/₹) costs: low ₹5 L, mid ₹15 L, high ₹30 L.
- **Digitization:** Scanning/OCR software licenses or contracts (e.g. ₹1 L/manuscript set). Low covers only volunteer scanning (₹2 L), mid includes some outsourced OCR (₹8 L), high budgets full digitization (₹20 L).
- **Travel/Workshops:** Conferences, collaborator meetings, user training. Low: ₹2 L; mid: ₹5 L; high: ₹10 L.
- **Outreach/Contingency:** Misc costs (community events, unexpected expenses). Low: ₹1 L; mid: ₹5 L; high: ₹10 L.

Table 2. Budget Breakdown (Total ₹L, 5-year, three scenarios)

Category	Low	Medium	High	Comments
Personnel (all staff)	80	150	250	Salaries + overheads
Tech/Infra (cloud)	25	50	80	Initial cloud hosting and storage; future intensive compute clusters (GPUs) will be supported by dedicated angel investment.
Digitization (scanning/O CR)	10	30	60	Hardware, software, contracts

Category	Low	Medium	High	Comments
Travel & Events	5	15	30	Conferences , workshops
Outreach & Misc	5	10	30	Marketing, documentation, contingency
Total	125	255	450	

(Figures in ₹ Lakhs; e.g. Medium total ~₹2.55 Cr.)

These estimates are illustrative. The **Medium** scenario (~₹2.5 Cr total) supports a lean but effective team, full development, and modest outreach. The **High** scenario (~₹4.5 Cr) could allow more staff, expanded courses, and international collaboration travel. The **Low** scenario relies heavily on volunteer labor and minimal travel. The proposal will seek funding in tranches (yearly grants or milestone-based) and will continually pursue additional funding (grants, patron gifts) to cover any shortfall.

Revenue & Sustainability

To sustain operations beyond initial grants, DLIIH will employ a mixed model. Notably, Phase 0 and Phase 1 will be funded primarily through a public crowdfunding and micro-donation campaign. The "Pre-Seed Heritage Campaign" will leverage established partnerships with prominent Twitter/X networks and Indocentric YouTube channels to secure immediate, agile funding for infrastructure and the Rigvedic MVP. Institutional grants are framed as a secondary focus reserved for long-term scaling.

- **Donations/Patrons:** We will solicit recurring donations (₹100–1000/month) via “Friends of DLIIH” programs. Large patrons (₹1–5 L/year) will get private reports and acknowledgments. This mirrors models of other Dharma projects (SuttaCentral, 84000) that sustain primarily on community support.
- **Institutional Licenses:** Academic libraries or universities may pay subscriptions (annual ₹50–200 k) for enhanced services (API access, usage analytics, premium data exports). Given unique corpus, many universities will find value in campus-wide licenses.
- **Premium Features:** A small “Pro” tier (₹5,000–10,000/year) could unlock advanced tools (bulk data export, API quotas, offline packages, AI research sandbox). Basic search and browsing remain free.
- **Educational Programs:** Offer fee-based online courses or certification in Vedic/Sanskrit reading through partner universities or via platforms like Coursera. Even modest fees (₹5–10k per course) for professional courses can add revenue. Scholarships ensure access.
- **Print-on-Demand Books:** Compile popular translations/commentaries into books sold via POD (cost-recovery plus margin).

- **Grants & Crowdfunding:** Continue to apply for research grants (Mellon, NEH, Ministry of Culture, numata, etc.) and periodic crowdfunding for special subprojects (e.g. “Sponsor a Sūtra” campaigns).
- **B2B Data Licensing:** A scalable revenue stream will be generated by licensing the platform’s clean, TEI-encoded, and morphologically tagged Vedic datasets to AI technology companies. Highly structured ancient language data is a rare, high-value asset essential for training specialized Indic Large Language Models (LLMs).
- **Angel Investment for Compute Infrastructure:** While Phase 0 and Phase 1 operational costs will be crowdfunded via digital mobilization partners, the project will seek targeted private/angel investment specifically for GPU and High-Performance Compute (HPC) clusters. Once the 'vertical slice' MVP proves the automated NLP pipeline for archaic Vedic grammar, private capital will be raised solely to secure the local GPU servers necessary to fine-tune Indic Large Language Models, ensuring data sovereignty and reducing long-term API costs.

No single source is assumed pre-committed; our sensitivity analysis (low/med/high budget) accounts for partial funding. By Year 3 we expect steady income from institutions and donations to cover operating costs, as seen in similar digital library initiatives.

Risk Analysis & Mitigation

- **Funding Shortfall:** *Mitigation:* Early seed grants (Year 0) and diversified funding as above; a conservative staffing plan allows downscaling if needed. Regular cost reviews will trim expenses.
- **Data Gaps/Licensing:** Most Vedic texts are PD; only modern translations carry copyright. We will use or negotiate CC licenses for translations. *Mitigation:* Legal review to avoid infringement; partner with institutions owning texts for permission (e.g. “Memory of World” texts).
- **Technical Complexity:** NLP in Sanskrit is hard (sandhi, inflection). *Mitigation:* Use existing Sanskrit NLP research (SanskritTagger【10†L207-L212】, Vyoma tools) and focus on user-friendly features first. Build modularly so components can improve over time.
- **Volunteer Reliance:** Quality may vary. *Mitigation:* Strong editorial QA (see workflow) and recognition (credits, co-authorship) to incentivize volunteers. Engage universities for credit/assignments.
- **Religious Sensitivity:** Some communities guard their traditions. *Mitigation:* Form a Cultural Advisory Board (scholars from relevant communities). Ensure neutrality (present multiple viewpoints). Include disclaimers (“scholarly translation only, not for rituals”).
- **Political/Institutional Risks:** Changes in government policy or institutional priorities could impact support. *Mitigation:* Keep project apolitical, emphasize academic/scientific framing. Build an international advisory board to broaden backing.

- **Volunteer/Donor Fatigue and Public Trust:** *Mitigation:* A "Proof of Progress" model involving radical transparency, including public GitHub code repositories, open financial reporting, and the real-time release of digitized, accented Rigvedic verses as milestones are achieved to maintain community enthusiasm.

Partnerships & Outreach

Strategic partners are crucial. We will collaborate with (prioritized list):

1. **Muktabodha IRI (USA/India):** Leading Sanskrit digital library with Vedic manuscript images【46†L55-L59】. Likely to provide data and expertise (e.g. their Gokarna Veda scans【13†L80-L88】) and join governance.
2. **Bhandarkar Oriental Research Institute (Pune):** UNESCO-acknowledged Rigveda manuscript collection【41†L251-L257】; we will coordinate digitization/metadata with their Manuscript Centre【41†L251-L257】.
3. **French Institute of Pondicherry (IFP, Puducherry):** Houses 2,000+ palm-leaf Sanskrit manuscripts (designated UNESCO MoW) including Vedic texts【13†L69-L77】【13†L80-L88】. We will arrange access to their catalog and scans.
4. **Göttingen Register (GRETIL):** For existing text corpora (mostly classical). We plan to harvest GRETIL's Vedic e-texts and enrich them with markup【10†L179-L187】 , giving mutual credit.
5. **Digital Corpus of Sanskrit (Heidelberg):** Team (SanskritTagger) for morphological tools and consulting on lexica search.
6. **IGNCA (India) & Gaudiya Grantha Mandira:** For content (e.g. Vaishnava Aranyakas, Upaniṣads) and outreach in scholastic communities.
7. **Universities & Institutes:** Partner with Sanskrit depts (e.g. Univ. of Mysore, U. of Hyderabad, JNU, SOAS, U. Chicago, Brown), promoting the portal to students for coursework and collaborative research.
8. **Religious/Cultural Institutions:** Temples (e.g. Kashi Viswanath, TTD), Vedic schools, and sabhās that might host manuscripts or encourage youth participation.
9. **Digital Mobilization & Media Partners:** This group includes our Twitter/X and YouTube collaborators, utilized for public project updates, the recruitment of vetted volunteer proofreaders, and driving transparent, milestone-based community fundraising.
10. **Tech Providers:** Engage IIT Kanpur (SANSKNET team), CDAC Pune (OCR expertise), and open-source communities (e.g. Apache, Linux Sanskrit) for technical support.

These partners will be approached with MOUs defining roles (data sharing, co-translation, user testing). An outreach plan includes academic workshops (to train scholars on the platform), webinars for practitioners, and social media updates. A

Partner Advisory Group will meet annually to align efforts.

Legal, IP & Ethical Considerations

- **Copyright:** The project will only use public-domain or openly licensed texts. Any in-copyright critical editions or translations will be used only with permission (and under CC BY-SA if possible). We will license all new content (translations, annotations) under a free license (e.g. CC BY-SA). The policy and license info will be prominently displayed【29†L369-L373】【53†L179-L184】.
- **Religious Sensitivity:** The DLIIH platform is strictly educational/scholarly. We will respect sectarian traditions by *documenting variations* (e.g. different Śākhās) and not prescribing any practice. Advisory committees including practitioners will vet content for potential offensiveness (e.g. in commentaries on gods).
- **Data Privacy:** No personal data is required from users. Contributors may register accounts; we will follow privacy norms and not share personal info.
- **Open Access:** In line with UNESCO and national mandates for cultural heritage, the library will be open-access. We will credit all contributors (scholars, donors, institutions) in metadata and publications.

Monitoring & Evaluation

Progress and impact will be measured by **Key Performance Indicators (KPIs)**:

- **Corpus growth:** Number of texts/verses encoded and online per quarter. Target: ≥200,000 Sanskrit words by Year 2; ≥1 million by Year 5.
- **Usage stats:** Monthly active users, search queries, API calls. Aim for steady growth (~100% YOY) and broad geography.
- **Scholarly output:** Citations of the library in academic works, co-publications, conference presentations (tracked via Google Scholar).
- **Community engagement:** Number of active contributors/volunteers, workshop attendees, courses completed.
- **Sustainability:** Funds raised vs. target, subscribers onboarded. Achievement of break-even operational budget by Year 4.

Regular reports will be produced (annually and at phase-gates) summarizing these metrics. A robust analytics backend will collect usage data (with user privacy in mind) to inform improvements. Feedback surveys (from scholars, students) will guide refinements. The Advisory Board will review evaluation reports to advise course corrections.

5-Year Roadmap & KPIs

By the end of Year 5, DLIIH aims to have:

- Digitized the complete Vedic corpus (all 4 Vedas + attached literature) with TEI encoding and English translations (≥1.5 million Sanskrit words)【41†L251-L257】【10†L190-L194】.

- Published 20+ scholarly articles or books (in translation series) based on the texts.
- Attracted a regular user base (e.g. >10,000 unique visits/month) and an active volunteer community (≥200 participants).
- Deployed advanced tools: a working semantic search and RAG assistant with archival of results.
- Secured sustainable funding (institutional licenses and donations covering ~70% of annual costs).

These outcomes will solidify the library as a global scholarly resource. The timeline (see above) ensures paced expansion, with reviews at each major milestone.

Visual aids: (a) A Mermaid timeline summarizing the 5-year plan (above). (b) An ER diagram (above) illustrating the data model for texts, editions, segments, translations, etc.

Sources: We have drawn on related projects and literature to inform this plan【4†L11-L19】【10†L179-L187】【46†L55-L59】【41†L251-L257】【53†L103-L111】. For example, the Sanskrit Library Integration project funded by NSF established key encoding standards【4†L11-L19】; Muktabodha’s digital library demonstrates large-scale Sanskrit corpus building【46†L55-L59】; and recent reviews identify the need for more semantic Vedic tagging【10†L179-L187】【53†L179-L184】. This proposal builds on such foundations to deliver an ambitious but achievable digital library of Indic thought.
